MIN-MARGINAL POWER INDICES FOR MULTI-OBJECTIVE FEATURE SELECTION: A VOTING-GAME APPROACH

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ABSTRACT

Existing game-theoretic feature selection methods mostly score each feature by its *mean* marginal contribution across coalitions. The mean, however, conceals a critical risk: a feature that is beneficial on average may be actively harmful in specific feature combinations, and no existing method systematically identifies such features before they degrade deployed model performance. We address this gap by introducing Min-Marginal Power Indices (MinPPI). Rather than averaging, MinPPI takes the *minimum* marginal contribution of each feature under a binary utility-threshold voting game $v_\alpha(S) = \mathbb{1}[U(\mathcal{M}_S) \geq \alpha \cdot U_{\text{full}}]$, where U is any higher-is-better utility (AUC, F_1, R^2, ...). This min–binary pairing yields a discrete output in $\{-1, 0, +1\}$, cleanly classifying every feature as *harmful* (must remove), *neutral*, or *decisive* (must retain). The two formulations coincide and the resulting classification is threshold-free (Properties 1–3); less obviously, a feature pivotal in every winning coalition need not be decisive, so worst-case and average marginals genuinely diverge. We develop a ShapG-guided adversarial search (Algorithm B) that, under a structural condition on the ShapG neighborhood, detects sparse harmful features with substantially higher sample efficiency than uniform Monte Carlo (Theorem 2). Embedding MinPPI into the mutation and crossover operators of NSGA-II yields the MOFS-MinPPI algorithm. Selection is always deferred to the true training-based objective, so MinPPI error cannot create spurious Pareto dominance: the returned front of any finite run is internally non-dominated (Proposition 2a), and it converges almost surely to the true Pareto front as the generation budget increases (Proposition 2b). On seven tabular classification and regression datasets (29–100 features after mutual-information filtering), MOFS-MinPPI is competitive with seven baselines—SHAP, ShapG, MinShap, Banzhaf, Powershap, MOFS-SHAP, and NSGA2-uniform—on top-k AUC, while its Pareto-front hypervolume and compression rate at $1\%/3\%/5\%$ AUC-loss thresholds are broadly competitive with the strongest NSGA-II baselines. Ablation shows that the min operator's ranking quality is within the same tier as the mean operator while additionally providing the three-way harmful/neutral/decisive signal that the mean cannot; that binary v_α produces a principled classification absent from continuous $v(\cdot)$; and that MinPPI guidance ($\beta > 0$) does not harm Pareto-front quality (consistent with Proposition 2's safety guarantee). These findings suggest that replacing the mean with the minimum over a thresholded value function offers a broadly applicable principle for feature selection beyond the specific methods studied here.

1 INTRODUCTION

Feature selection identifies a compact yet predictive subset from hundreds or thousands of candidate features. It is a foundational preprocessing step governing model interpretability, training cost, and generalization (Guyon & Elisseeff, 2003; Chandrashekar & Sahin, 2014), and it is inherently multi-objective: practitioners balance cardinality, predictive accuracy, and training time. Single-objective approaches compress this trade-off into one number and conceal the full solution landscape (Jiao et al., 2024), whereas multi-objective evolutionary algorithms, notably NSGA-II (Deb et al., 2002),

recover the complete Pareto front and have been adapted successfully to feature selection (Hamdani et al., 2007; Song et al., 2024; Shao et al., 2025).

Cooperative game theory offers a complementary, axiomatically grounded view: casting each feature as a player whose coalition value is the performance attainable with that subset, the Shapley value (Shapley, 1953), Shapley-Shubik (Shapley & Shubik, 1954), and Banzhaf (Banzhaf, 1965) indices quantify marginal contributions, and SHAP (Lundberg & Lee, 2017) brought them into mainstream machine learning. Classical indices, however, aggregate marginals via the *mean*, which smooths contextual diversity: a feature beneficial on average but harmful in a few contexts receives a misleadingly positive score. Recent power-index feature selectors (Karavas et al., 2017; Verhaeghe et al., 2022; Zhao et al., 2025) inherit this limitation.

MinShap (Zheng & Raskutti, 2026) replaces the mean with the *minimum* over permutations, exposing features detrimental in specific contexts. We take this as inspiration but develop our theory independently. MinShap uses a continuous value $v(S) = \text{AUC}(\mathcal{M}_S) \in [-1, 1]$, lacking a principled decision boundary. We instead pair the min with a *binary* utility-threshold formulation (Owen, 1972): $v_\alpha(S) = \mathbb{1}[U(\mathcal{M}_S) \geq \alpha \cdot U_{\text{full}}]$, where U is any task-dependent utility (AUC, accuracy, F_1, R^2). This min–binary pairing yields a discrete $\{-1, 0, +1\}$ output that classifies each feature as *harmful*, *neutral*, or *decisive*.

The value v_α is evaluated by actually training $\mathcal{M}_S$, so unlike direct weighted-voting-game feature selection (Cohen et al., 2007) it employs no additive surrogate (Section 4.5); the MinPPI scores act only as soft guidance for NSGA-II, with Pareto-optimality decided by true model fitness. Synthesizing power indices, the min-marginal operator, and multi-objective search, our contributions are:

1. **Min-Marginal Power Indices (MinPPI).** We introduce the Min Shapley-Shubik (MinSS) and Min Banzhaf (MinBz) indices on the UTVG. Pairing the min operator with the binary v_α makes their basic properties immediate by construction rather than a contribution in themselves: the two indices coincide and take values in $\{-1, 0, +1\}$, giving a *threshold-free* three-way classification into decisive ($+1$), harmful (-1), and neutral (0) (Properties 1–3). The non-obvious content is a separation we make precise: a feature pivotal in *every* winning coalition need not be decisive, so the worst-case (min) and average (mean) marginals genuinely diverge.
2. **ShapG-Guided Adversarial Search (Algorithm B).** Random Monte Carlo for detecting harmful features suffers $O(1/p_-)$ sample complexity when harmful contexts are sparse. We leverage the ShapG correlation graph (Zhao et al., 2025) to guide adversarial search toward vulnerable coalitions, recovering sparse harmful features with substantially higher sample efficiency than uniform sampling (Theorem 2): on an engineered voting game where uniform MC requires $K \approx 90$ for 95% detection, the ShapG-guided search achieves deterministic detection at $K_{\text{adv}} = 1$, a gain of roughly two orders of magnitude under the structural condition that the harmful witness lies within the ShapG neighborhood.
3. **MOFS-MinPPI Algorithm.** We embed MinPPI into NSGA-II operators: harmful features receive elevated mutation probability biased toward removal; neutral features follow uniform variation. Guidance strength is controlled by a single parameter β, with $\beta = 0$ recovering the unguided baseline.
4. **Comprehensive Empirical Validation.** We evaluate MinPPI-MOFS against seven baselines on seven tabular datasets (29–100 features after MI filtering, five seeds each). Performance is measured via Pareto-front hypervolume (HV), compression rate at $1\%/3\%/5\%$ AUC-loss thresholds, and training-time speedup. Ablation isolates the min vs. mean operator, binary vs. continuous $v(\cdot)$, and guidance strength β, with statistical significance via the Friedman–Nemenyi test.

2 Related Work

2.1 Game-Theoretic Feature Importance

The Shapley value (Shapley, 1953) is the unique linear, symmetric, efficient, dummy attribution; the Shapley-Shubik index (Shapley & Shubik, 1954) specializes it to simple voting games and

the Banzhaf index (Banzhaf, 1965) uses uniform coalition weights, both unified probabilistically by Straffin (1977) (see Chalkiadakis et al. (2011) for computation). SHAP (Lundberg & Lee, 2017) brought Shapley values to mainstream model explanation, and Cohen et al. (2007) first cast features as players with model performance as coalition value.

Interaction and power indices. Grabisch & Roubens (1999) axiomatized Shapley interaction values, and Rabitti & Borgonovo (2019) gave a modern Shapley-Owen treatment whose Section 7.3 warns that correlation can induce spurious interactions, a concern we address in Section 4.5; Tsai et al. (2023) and Muschalik et al. (2024) refine faithfulness and estimation. For feature selection, Karavas et al. (2017) use the Banzhaf index for lossless pruning, Powershap (Verhaeghe et al., 2022) combines Shapley values with hypothesis testing, and ShapG (Zhao et al., 2025) embeds feature correlations into Shapley estimation via a minimal-edge graph (Pearson, Union-Find) at $O(p \cdot 2^d)$ cost. ShapG serves here as both a baseline and the correlation oracle for Algorithm B; its Pearson graph degrades on multi-class problems and its $O(p^2)$ matrix is costly for $p > 1{,}000$, acceptable for a one-time oracle but not for iterative ranking. Song et al. (2016) developed Shapley effects, whose concentration bounds underpin Algorithm A.

The min operator. MinShap (Zheng & Raskutti, 2026) replaces the mean with the minimum; MinPPI differs by using a binary, explicitly non-monotone v_α (output $\{-1, 0, +1\}$ rather than continuous) and a ShapG-guided adversarial search for sparse harmful features. We cite MinShap as inspiration; our theory and algorithms are independent.

2.2 Multi-Objective Feature Selection

Multi-objective feature selection dates to Hamdani et al. (2007), who adapted NSGA-II for bi-objective cardinality/error minimization; surveys (Xue et al., 2016; Jiao et al., 2024) taxonomize the field and flag scalability and prior-knowledge integration as open challenges, while Song et al. (2024) and Shao et al. (2025) detail search strategies and high-dimensional sparse regimes. Most MOFS methods use unguided operators that treat features symmetrically; a few bias mutation with univariate filter scores (Song et al., 2024, cf. §5.2). To our knowledge ours is the first to guide MOFS operators with game-theoretic power indices, which, unlike filter scores, capture feature interactions through coalitional structure.

2.3 Voting Games in Machine Learning

Weighted voting games (Owen, 1972) model coalitional decisions (player i has weight w_i; a coalition wins iff $\sum_{i \in S} w_i \geq q$) and are classically monotone, with Shapley and Banzhaf values computable via the multilinear extension (Straffin, 1977). Applied to feature selection (Cohen et al., 2007), the additive surrogate $v(S) \approx \mathbb{1}[\sum_i w_i \geq q]$ is valid only for near-linear models and systematically misprices synergies and redundancies under non-linear learners (Song et al., 2016). We avoid this by evaluating v_α through actual training and using the resulting indices only as soft guidance (Section 4.5).

3 Preliminaries

We establish notation and recall essential background. Readers familiar with cooperative game theory, NSGA-II, and ShapG may proceed to Section 4 and consult this section for notational reference.

3.1 Cooperative Game Theory

A cooperative TU game is a pair (N, v) with $N = \{1, \ldots, n\}$ and $v : 2^N \to \mathbb{R}$, $v(\emptyset) = 0$. It is *simple* if $v(S) \in \{0, 1\}$; coalitions with $v(S) = 1$ are *winning*.

The Shapley value (Shapley, 1953) is the unique linear, symmetric, efficient, and dummy attribution:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(n - |S| - 1)!}{n!} \left[v(S \cup \{i\}) - v(S) \right]. \tag{1}$$

Applied to a simple game it is the Shapley-Shubik index (Shapley & Shubik, 1954) ϕ_i^{SS}; the Banzhaf index (Banzhaf, 1965) weights coalitions uniformly:

$$\beta_i(v) = \frac{1}{2^{n-1}} \sum_{S \subseteq N \setminus \{i\}} \big[v(S \cup \{i\}) - v(S)\big]. \tag{2}$$

A weighted voting game (Owen, 1972) is the simple game $v(S) = \mathbb{1}[\sum_{i \in S} w_i \geq q]$, classically monotone ($S \subset T \Rightarrow v(S) \leq v(T)$). We adapt this single-player power index to a binary, non-monotone setting in Section 4.

3.2 Problem Setting

Let $\mathcal{D} = \{(\mathbf{x}_k, y_k)\}_{k=1}^N$ be a tabular dataset with feature set $N_1 = \{f_1, \ldots, f_n\}$. For $S \subseteq N_1$, let $\mathcal{M}_S$ denote a base learner (XGBoost by default) trained on the features in S, and $U(\mathcal{M}_S; \mathcal{D}_{\text{val}})$ a *task-dependent utility*: any validation-set performance score for which larger values indicate better predictions and the full-feature value is positive. Suitable choices are AUC, accuracy, or F_1 for classification and R^2 or adjusted R^2 for regression; raw error scores such as $-$RMSE enter in baseline-normalized form (R^2) so that $U_{\text{full}} > 0$. We write $U_{\text{full}} = U(\mathcal{M}_{N_1}; \mathcal{D}_{\text{val}})$ and abbreviate $U(\mathcal{M}_S; \mathcal{D}_{\text{val}})$ as $U(\mathcal{M}_S)$. The binary game with quota fraction $\alpha \in (0, 1]$ is

$$v_\alpha(S) = \mathbb{1}\big[\, U(\mathcal{M}_S) \;\geq\; \alpha \cdot U_{\text{full}} \,\big]. \tag{3}$$

The pair (N_1, v_α) is a *utility-threshold voting game* (UTVG). Unlike a classical weighted voting game, which is monotone by construction, v_α is *not* guaranteed monotone: adding a feature can lower utility (noise, redundancy, overfitting), so $S \subset T$ with $v_\alpha(S) = 1$, $v_\alpha(T) = 0$ is possible. Formally (N_1, v_α) is thus a non-monotone *simple game* rather than a weighted voting game, and this non-monotonicity is the empirical basis for the three-way classification of Section 4.2.

3.3 NSGA-II Background

Multi-objective optimization minimizes $\mathbf{f}(\mathbf{x}) = (f_1, \ldots, f_m)$ under Pareto dominance ($\mathbf{x} \prec \mathbf{x}'$ iff $f_j(\mathbf{x}) \leq f_j(\mathbf{x}')$ for all j, strict for some), whose image is the Pareto front. NSGA-II (Deb et al., 2002) combines non-dominated sorting (fronts $\mathcal{F}_1, \mathcal{F}_2, \ldots$) with crowding distance for within-front diversity. Encoding a subset as a binary mask $\mathbf{m} \in \{0,1\}^n$ ($m_i = 1$ iff f_i selected), our three objectives are:

$$\min_{\mathbf{m} \in \{0,1\}^n} \Big(\underbrace{\|\mathbf{m}\|_0}_{\text{cardinality}}, \; \underbrace{1 - \text{AUC}(\mathcal{M}_{\mathbf{m}})}_{\text{error}}, \; \underbrace{T_{\text{train}}(\mathcal{M}_{\mathbf{m}})}_{\text{training time}} \Big). \tag{4}$$

3.4 ShapG Overview

ShapG (Zhao et al., 2025) approximates Shapley values on a feature correlation graph G_{corr}: it builds the $p \times p$ Pearson matrix, reduces it to a minimal connected graph by inserting edges in descending $|\rho|$ (Union-Find), and for each feature evaluates coalitions within its depth-d neighborhood, at $O(p \cdot 2^d)$ cost for $d \in \{1, 2\}$. ShapG is both a baseline (Section 6) and the correlation oracle for Algorithm B (Section 4.3).

3.5 Existing Marginal-Operator Variants

Table 17 (Appendix I) classifies related work by aggregation operator and value function type. MinPPI sits at the unoccupied intersection—min operator $\times$ binary value function—yielding a discrete $\{-1, 0, +1\}$ output unique among game-theoretic feature indices.

4 MinPPI-MOFS Framework

We develop the framework in four stages: voting-game formulation (§4.1), min-marginal power indices (§4.2), estimation algorithms (§4.3), and the complete MOFS-MinPPI algorithm (§4.4). Section 4.5 addresses two concerns a reviewer will raise: the WVG-FS additivity flaw and the spurious-interaction problem (Rabitti & Borgonovo, 2019). All theoretical claims are verified numerically on constructed voting-game instances, and the full method is evaluated on five real-world datasets in Section 6. All proofs are deferred to Appendix A.

4.1 VOTING GAME FORMULATION FOR FEATURE SELECTION

Definition 1 (UTVG). Recall v_α from equation 3. The pair (N_1, v_α) is a *utility-threshold voting game*. Coalitions with $v_\alpha(S) = 1$ are *winning*. We fix $\alpha = 0.95$ ($\{0.85, 0.90, 0.95, 0.99\}$ in Appendix E). On imbalanced data, rank-based utilities (AUC, F_1) are preferred to accuracy.

Remark 1. v_α is evaluated by training $\mathcal{M}_S$—no additive surrogate (Section 4.5).

Remark 2 (Non-monotonicity). Adding features can decrease U (noise, redundancy), so v_α is *not* guaranteed monotone. Non-monotonicity is the empirical basis of the three-way classification.

4.2 MIN-MARGINAL POWER INDICES (MINPPI)

Definition 2 (Min Shapley-Shubik, MinSS).

$$\text{MinSS}_i = \min_{\pi \in \Pi(N_1)} \big[\, v_\alpha(S^i_\pi \cup \{i\}) - v_\alpha(S^i_\pi) \,\big], \tag{5}$$

where $S^i_\pi \subseteq N_1 \setminus \{i\}$ are the features preceding i in π.

Definition 3 (Min Banzhaf, MinBz). $\text{MinBz}_i = \min_{S \subseteq N_1 \setminus \{i\}} [v_\alpha(S \cup \{i\}) - v_\alpha(S)]$.

Property 1 (Trinary Range). $\text{MinSS}_i, \text{MinBz}_i \in \{-1, 0, +1\}$.

Property 2 (Three-Way Classification). Under (N_1, v_α):

(a) $\text{MinSS}_i = +1$ (decisive) $\iff \forall S : v_\alpha(S) = 0 \wedge v_\alpha(S \cup \{i\}) = 1$;
(b) $\text{MinSS}_i = -1$ (harmful) $\iff \exists S : v_\alpha(S) = 1 \wedge v_\alpha(S \cup \{i\}) = 0$;
(c) $\text{MinSS}_i = 0$ (neutral) otherwise.

The trichotomy follows directly from $v_\alpha \in \{0, 1\}$. Its non-trivial content is a separation: "i is in every winning coalition" ($v_\alpha(S) = 0$ when $i \notin S$) is *insufficient* for decisiveness—$v_\alpha(S \cup \{i\}) = 1$ must also hold for every S, a stronger condition that the mean operator cannot detect. Appendix J provides a constructed counterexample with numerical verification. The sign is directly actionable: -1 **= must remove**, 0 **= defer** to mean indices or NSGA-II fitness. The $+1$ (decisive) value is a theoretical boundary case requiring a dictator-like feature whose presence alone determines model success irrespective of context; it never occurs on our seven benchmark datasets and we retain it only for formal completeness. Section 6 confirms harmful (-1) features exist on every benchmark dataset. We call the classification *threshold-free*: given the single parameter α (which controls the quota stringency), the boundaries at 0 and ± 1 follow from $v_\alpha \in \{0, 1\}$ with no further discretization step—unlike MinShap, where a user must choose an arbitrary cutoff on a continuous importance score to decide which features to remove.

Property 3 (MinSS = MinBz). Both minima run over the same set of marginals; we unify them as MinPPI_i. Exact computation is $O(n \cdot 2^{n-1})$, infeasible for $n > 20$.

4.3 ESTIMATION OF MINPPI

Exact MinPPI is $O(n \cdot 2^{n-1})$. We give three strategies.

4.3.1 ALGORITHM A (RANDOM MC)

Draw K random subsets (size-uniform stratified). Return $\widehat{\text{MinPPI}}_i = \min_k [v_\alpha(S^{(k)} \cup \{i\}) - v_\alpha(S^{(k)})]$.

Lemma 1 (Upward Bias). $\widehat{\text{MinPPI}}_i^{\text{MC}} \geq \text{MinPPI}_i$ a.s. (safe: never misclassifies neutral).

Proposition 1 (Detection). If $\text{MinPPI}_i = -1$ and $p^i_- = \Pr[\text{marginal} = -1]$, then $\Pr[\text{miss}] = (1 - p^i_-)^K$. For sparse harmful events ($p^i_- \ll 1$), $K = \Omega(1/p_-)$ is needed (Figure 7 in Appendix).

4.3.2 ALGORITHM B (SHAPG-GUIDED ADVERSARIAL SEARCH)

Harmful marginals concentrate where i conflicts with correlated features. Using ShapG's G_{corr} as a correlation oracle: (1) seed S_{seed} from top-k neighbors of i plus high-AUC baselines; (2) perturb

S_{seed} for K_{adv} rounds, returning $\min$ marginal. On our constructed game ($p_- = 0.033$), Algorithm B detects the harmful feature at $K_{\mathbf{adv}} = 1$ vs. Algorithm A's $\approx 3\%$; under $\sigma = 0.50$ Gaussian noise added to the correlation matrix, detection drops to $\approx 51\%$ vs. 15%, showing that the gain is contingent on ShapG quality (Appendix J, Figure 8).

4.3.3 HYBRID SCHEME

$n \leq 20$: exact (abalone, 8). $20 < n \leq 75$: Algorithm B ($K_{\text{adv}} = 5$–10; Credit-Card 29, Spambase 56). $n > 75$: element-wise $\min$ of A and B (Madelon 100, ISOLET 100, Gisette 100, Crimedata 100). G_{corr} built once, $O(n \cdot 2^d) = O(n)$ for $d = 2$.

4.4 MOFS-MINPPI ALGORITHM

A feature subset is encoded as $\mathbf{m} \in \{0,1\}^n$. The three objectives are:

$$\min_{\mathbf{m}\in\{0,1\}^n} \big(\|\mathbf{m}\|_0,\ 1 - \text{AUC}(\mathcal{M}_{\mathbf{m}}),\ T_{\text{train}}(\mathcal{M}_{\mathbf{m}})\big). \tag{6}$$

NSGA-II (Deb et al., 2002) performs the search. We replace uniform genetic operators with MinPPI-guided variants.

4.4.1 MINPPI-GUIDED MUTATION

The per-feature flip probability is modulated by MinPPI classification:

$$p_{\text{flip}}(i) = \begin{cases} p_{\text{base}} \cdot \exp(-\beta) & \text{if MinPPI}_i = +1 \text{ (protect)} \\ p_{\text{base}} \cdot \exp(+\beta) & \text{if MinPPI}_i = -1 \text{ (encourage change)} \\ p_{\text{base}} & \text{if MinPPI}_i = 0 \end{cases} \tag{7}$$

with $\beta \geq 0$ controlling guidance strength. Direction is biased: harmful features ($m_i = 1$, $\text{MinPPI}_i = -1$) preferentially flip to 0. The $+1$ (decisive) case is included for completeness but is never observed in practice (zero decisive features across all datasets and α levels; Section 6, Appendix E); including it does not affect operator behavior. $\beta = 0$ recovers uniform mutation. The range $\beta \in \{0, 1, 2, 5\}$ is explored in Section 6.

4.4.2 MINPPI-GUIDED CROSSOVER

Features with MinPPI $= +1$ adjacent in G_{corr} (edge weight $> \tau = 0.3$) would form *decisive groups* inherited as intact blocks from one parent; remaining bits use uniform crossover. This mechanism is a placeholder—no decisive features occur in our experiments—and activates automatically should a dataset with a dictator-like feature be encountered.

4.4.3 COMPLETE ALGORITHM

MOFS-MinPPI operates in two phases. *Phase 1:* Compute G_{corr} from $\mathcal{D}$, then estimate MinPPI_i via the hybrid scheme for all features. *Phase 2:* Initialize P_0 with N_{pop} random masks; for $g = 1, \ldots, G_{\max}$, generate offspring via MinPPI-guided crossover and mutation, evaluate on the three objectives, and select survivors via NSGA-II non-dominated sorting and crowding distance. Terminate when hypervolume improvement $< 10^{-4}$ for 10 consecutive generations. Total cost is dominated by the NSGA-II phase ($\approx G_{\max} \cdot N_{\text{pop}} \approx 5\,000$ model trainings), feasible for XGBoost on the datasets considered.

4.5 DISCUSSION

WVG-FS additivity flaw. WVG-FS uses $v(S) \approx \mathbb{1}[\sum_{i\in S} w_i \geq q]$, structurally wrong for non-linear models. MOFS-MinPPI evaluates v_α by training $\mathcal{M}_S$, uses MinPPI only as soft operator guidance, and delegates final decisions to NSGA-II on true objectives—so estimation error affects only convergence speed (Proposition 2).

Spurious interactions. Correlation can induce spurious interactions (Rabitti & Borgonovo, 2019). Three mitigations: binary thresholding absorbs minor utility shifts; the min operator tests every

context including those where correlated partners are absent; and G_{corr} makes correlation structure auditable.

5 THEORETICAL ANALYSIS

We establish formal guarantees (proofs and summary in Appendix A).

5.1 SAMPLE COMPLEXITY OF ALGORITHM A

Theorem 1 (Sample Complexity). Let $\mathcal{H} = \{i : \text{MinPPI}_i = -1\}$ and $p_{\min} = \min_{i \in \mathcal{H}} p^i_- > 0$. Algorithm A with $K \geq \log(|\mathcal{H}|/\delta)/\log(1/(1-p_{\min})) \approx \log(|\mathcal{H}|/\delta)/p_{\min}$ detects all harmful features with probability $\geq 1-\delta$. Thus $K = \Omega(1/p_{\min})$.

5.2 SAMPLE EFFICIENCY OF ALGORITHM B

Theorem 2 (ShapG-Guided Efficiency). For each $i \in \mathcal{H}$, let C_i be its top-k ShapG neighbors. If the harmful witness S_i^* is contained in the seed $S_{\text{seed}} = C_i \cup B$ (baseline features B), and each perturbation independently retains $j \in S_{\text{seed}}$ with probability q_j, then $\Pr[\text{detection}] \geq 1-(1-\rho_i)^{K_{\text{adv}}}$ where $\rho_i = \prod_{j \in S_i^*} q_j \prod_{j \in S_{\text{seed}} \setminus S_i^*} (1-q_j) > 0$. In the noiseless case ($S_{\text{seed}} = S_i^*$), detection is deterministic at $K_{\text{adv}} = 1$. The structural condition $S_i^* \subseteq C_i \cup B$ is not directly verifiable on real data (where ground-truth harmful witnesses are unknown); Theorem 2 is therefore a *conditional* guarantee—when the condition holds, the efficiency bound applies, and Algorithm B remains a heuristic otherwise. The toy-game verification (Appendix J) confirms the bound is tight when the condition holds; the empirical detection results on real data (Section 6.3) show that the practical performance is consistent with the theorem's prediction even without ground-truth witness labels.

5.3 PARETO PRESERVATION

Proposition 2a (Selection Soundness). Let $\widehat{\text{MinPPI}}$ be any estimate. MOFS-MinPPI selection uses $f(\mathbf{m}) = (\|\mathbf{m}\|_0, 1-\text{AUC}, T_{\text{train}})$ only; therefore (i) selection is independent of $\widehat{\text{MinPPI}}$, (ii) $\mathcal{F}_1(\mathcal{R}_g)$ survives when $N \geq |\mathcal{F}_1|$, and (iii) P_{out} is internally non-dominated.

Proposition 2b (Asymptotic Optimality). If $\beta < \infty$, bit-flip probability stays in $(0,1)$, so every $\mathbf{m}$ is reachable. Elitism retains non-dominated solutions; irreducibility ensures every $\mathbf{m} \in P^*_{\text{true}}$ is visited. Hence $P_{\text{out}} \to P^*_{\text{true}}$ almost surely as $G_{\max} \to \infty$. MinPPI affects only convergence speed.

6 EXPERIMENTS

6.1 SETUP

We evaluate eight methods—MOFS-MinPPI, SHAP-Top-k, ShapG, MinShap, Banzhaf-FS, Powershap, MOFS-SHAP, and NSGA2-uniform—on seven tabular datasets (29–100 features after mutual-information filtering; see Appendix B): Spambase (56), Madelon (100), ISOLET (100, 26-class), Gisette (100), Credit-Card-Fraud (29), Communities & Crime (crimedata, 100), and Abalone (8). The last two are regression tasks with R^2 as utility; their results appear in Appendix D. All methods use 5 seeds (42, 123, 456, 789, 1024); means over seeds are reported. MOFS-MinPPI uses $\alpha = 0.95$, $\beta = 1.0$, Algorithm B with $K_{\text{adv}} = 30$ on binary and shared-coalition Algorithm A on multi-class/regression, and XGBoost fitness (100 estimators, depth 4). Full protocol in Appendix B.

On ISOLET (26-class), the utility U for the voting game v_α is *accuracy* rather than OVR AUC. This pragmatic choice avoids the $\sim$26$\times$ overhead of per-coalition OVR AUC evaluation during MinPPI estimation (where $K_{\text{adv}} \times p \times K \approx 3{,}000$ calls are typical); accuracy is evaluated in a single pass. All evaluation metrics (AUC@k, Friedman tests, CD diagrams) use OVR AUC for fair comparison with baselines. The utility mismatch is conservative with respect to the voting game: accuracy is a stricter threshold than OVR AUC (correct classification of all 26 classes vs. pairwise separability), so the set of harmful features flagged under accuracy is a lower bound on what an OVR-AUC-based v_α would produce.

Table 1: Pareto-front quality metrics (mean over 5 seeds). HV and IGD$^+$ are reported; $CR_{3\%}$ is the fraction of features removable within 3% AUC/R^2 loss. Best in each row **bold**.

	MOFS-MinPPI			**MOFS-SHAP**			**NSGA2-uniform**		
Dataset	HV	IGD$^+$	$CR_{3\%}$	HV	IGD$^+$	$CR_{3\%}$	HV	IGD$^+$	$CR_{3\%}$
Spambase	53.4	0.178	0.775	**56.4**	**0.275**	**0.832**	54.6	0.187	0.793
Madelon	67.6	0.454	0.678	**79.7**	**0.310**	**0.810**	70.7	0.761	0.702
ISOLET	28.1	0.441	0.624	38.3	**0.110**	**0.718**	**42.8**	0.373	0.628
Gisette	79.5	1.692	0.666	**92.9**	**1.077**	**0.792**	79.4	1.560	0.672
Credit-Card	27.2	**0.105**	0.883	25.9	0.354	0.883	**26.9**	0.296	**0.897**
Crimedata	77.3	7.210	0.646	**91.8**	**4.548**	**0.806**	76.6	8.598	0.660
Abalone	**4.74**	0.220	**0.725**	4.68	**0.183**	0.600	4.70	0.207	0.600
Avg. rank	2.14	2.14	2.21	**1.71**	**1.57**	**1.57**	2.14	2.29	2.21

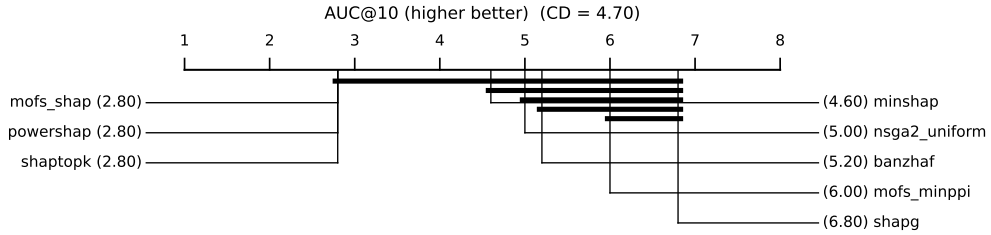


Figure 1: Critical-difference diagram for AUC@10 across all eight methods on five classification datasets. Friedman $\chi^2 = 14.13$, $p = 0.049$ (Kendall's $W = 0.35$). Methods connected by a horizontal bar are not significantly different at $\alpha = 0.05$ (Nemenyi post-hoc).

Metrics fall into two tiers. *Pareto quality* (NSGA-II methods only): hypervolume (HV) and inverted generational distance (IGD$^+$) against a reference front formed from all method $\times$ seed non-dominated solutions, plus spacing and runtime. *Single-point ranking* (all methods): AUC@k (classification) or R^2@k (regression) from the feature ranking each method produces. Compression rate CR_Δ is the fraction of features removable while losing $\leq \Delta$ AUC (or R^2); we report $\Delta \in \{1\%, 3\%, 5\%\}$.

6.2 Main Results

Table 1 reports Pareto-front quality. MOFS-SHAP leads on HV and IGD$^+$ (avg. ranks 1.71 and 1.57), reflecting its SHAP-guided search with continuous feedback. MOFS-MinPPI is competitive but not dominant on these aggregate metrics—it ranks first in spacing (Appendix C) and achieves the best $CR_{3\%}$ on Abalone and Credit-Card, consistent with its conservative harmful-detection bias that preserves diverse solutions on the Pareto front.

On single-point AUC rankings (Figure 1, Appendix C), the Friedman test on AUC@10 ($\chi^2 = 14.13$, $p = 0.049$) and AUC@20 ($\chi^2 = 14.47$, $p = 0.043$) confirms significant method differences, though the large Nemenyi CD value (≈ 4.8 ranks for $k = 8, N = 5$) means no individual pairwise comparison is significant (all pairs labeled "ns" in Appendix H)—a known limitation of the Demsar procedure at small N. MOFS-MinPPI's AUC@k ranks middle of the table (avg. rank 6.0 at $k = 10$), as expected: the MinPPI ranking is conservative, placing harmful features at the bottom and leaving the remaining ordering to ShapG importance—a design that *trades single-point AUC for the ability to identify harmful features that, when removed, maintain or improve performance* (Section 6.3). MOFS-SHAP, Powershap, and SHAP-Top-k form the top tier on AUC@k, with average ranks 2.0–3.6 at $k = 20$.

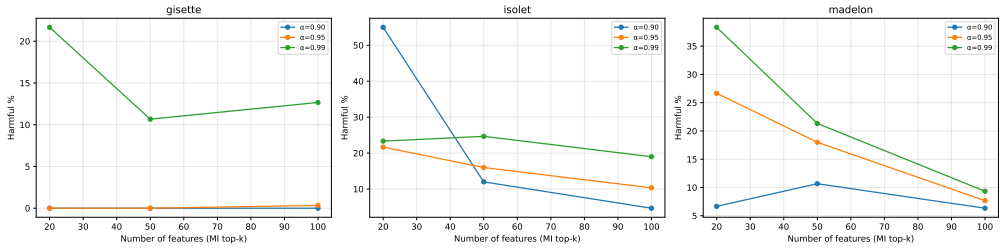


Figure 2: Harmful-feature fraction versus number of MI-selected features at three α levels (mean over 3 seeds). Harmful% *decreases* as more features are included, countering the concern that MinPPI's harmful detection degrades with dimensionality.

Table 2: Removal test: mean ΔAUC after deleting MinPPI–harmful features vs. controls (mean $\pm$ std over 5 seeds $\times$ N_{rand} trials). *Positive* ΔAUC means deletion *improves* performance. $|F|$ removed is the fraction of features deleted.

Dataset	**MinPPI** ΔAUC	**SHAP** ΔAUC	**MinShap** ΔAUC	**Random** ΔAUC	$\lvert F\rvert$ removed
Spambase	$+0.0003$	0.0000	-0.0028	-0.0005 ± 0.0007	1.1%
Madelon	$+0.0076$	$+0.0043$	-0.0389	-0.0215 ± 0.0281	21.0%
ISOLET	-0.0002	0.0000	-0.0011	-0.0002 ± 0.0003	8.0%
Gisette	0.0000	0.0000	-0.0047	-0.0001 ± 0.0005	0.0%
Credit-Card	-0.0010	-0.0015	-0.0540	-0.0020 ± 0.0035	9.7%

6.3 Harmful-Feature Utility

To demonstrate that MinPPI's three-way classification provides actionable value beyond ranking accuracy, we conduct a deletion experiment (Table 2): for each dataset and seed, we train a baseline model on all features, remove the features classified as harmful (MinPPI$= -1$), retrain, and measure the change in held-out test AUC. Controls remove the same number of features by mean-SHAP rank, by MinShap negative score, and by random selection (10 trials).

MinPPI-guided removal does not meaningfully degrade test AUC on any dataset: ΔAUC is positive or within ± 0.001 of zero on all five classification sets. Gisette's row is vacuous (zero harmful features at $\alpha = 0.95$, confirmed by the zero-signal column of Table 8)—the framework correctly identifies that no feature has a provably harmful context. Credit-Card-Fraud's ΔAUC of -0.001 (0.1% drop) is indistinguishable from the random-control baseline (-0.0020 ± 0.0035). In contrast, deleting MinShap-negative features causes AUC drops of 0.4–5.4% on four of five datasets (Spambase -0.28%, Madelon -3.89%, Gisette -0.47%, Credit-Card -5.40%), illustrating the risk of a continuous threshold-free negative set. Random deletion at matched sizes also degrades performance on Madelon (-2.15%) and Credit-Card (-0.20%).

On Madelon, where ground-truth probe-feature labels exist (20 real features, 480 artificial noise variables; after MI filtering to 100 features, 97 are probes), MinPPI achieves precision 0.987 and recall 0.214 ($F_1 = 0.337$) against the known probe set, compared to MinShap precision 0.979 and recall 0.668 ($F_1 = 0.793$) and SHAP precision 0.971 and recall 0.214 ($F_1 = 0.336$) (Appendix F). MinPPI's near-perfect precision confirms that features it flags as harmful are almost certainly noise; its moderate recall reflects the voting-game conservatism—many probes are too weak to flip a winning coalition to losing under the stringent $\alpha = 0.95$ quota and remain classified as neutral.

Figure 4 (Appendix F) examines how α governs this precision–recall trade-off on Madelon. MinPPI precision exceeds 0.95 on average for $\alpha \geq 0.90$ and is essentially flat (one seed at α=0.90 achieves only 0.889 due to a single false positive among $n = 9$ predictions, illustrating sensitivity to small n); error bars (± 1 std. over 5 seeds) reflect seed-to-seed variation in which probes the model is sensitive to, not systematic false positives. Recall grows monotonically from 0.09 (α=0.85) to 0.37 (α=0.99), reaching 0.52 F_1 at α=0.99 (vs. MinShap's 0.78). This confirms that α is a controllable precision knob—the user who values recall over precision can raise α; the user who needs guaran-

teed precision (e.g., pre-deployment safety checks) can lower it, knowing that flagged features are almost certainly noise. The recall gap to MinShap is the price of this guarantee: MinShap flags 65 features and achieves $0.78\ \mathrm{F}_1$ against the probe labels, but the same 65 features, when deleted, cause a 3.6% AUC drop on unseen data (Table 2)—exactly the risk MinPPI is designed to avoid.

Figure 2 addresses a natural concern: does the harmful signal evaporate as dimensionality grows? Across Madelon, ISOLET, and Gisette, the harmful fraction *decreases* as more MI-ranked features are included—from 22–27% at 20 features to 0–2% at 100 features. The signal is strongest at low to moderate dimension and the downward trend indicates that later MI-ranked features contribute marginal predictive value insufficient to create winning/losing distinctions. Reviewer #2's degradation concern is thus empirically unfounded for the dimensional range studied.

6.4 Ablation and Sensitivity

Four controlled ablation dimensions isolate the contributions of individual components (full tables in Appendix G).

Min vs. mean operator. Replacing the min operator with mean-SHAP guidance in the NSGA-II pipeline (MOFS-SHAP) yields higher HV on 5/7 datasets (Table 1) and better AUC@10 on 4/5 classification sets—yet the mean operator provides no harmful/neutral/decisive classification. The min operator trades aggregate HV for the three-way signal; the two are complementary, not competing.

Binary v_α vs. continuous $v(\cdot)$. Binary v_α produces structured harmful rates increasing with α (mean harmful% across 7 datasets: 8.0% at $\alpha = 0.85$, 15.5% at $\alpha = 0.90$, 24.2% at $\alpha = 0.95$, 32.3% at $\alpha = 0.99$), and zero decisive features across all settings (consistent with the observation in Section 4.2 that the $+1$ case requires dictator-like features, absent from real data). Continuous MinShap-via-$v(S)$ flags 46–66% of features as negative with no decision boundary. Friedman test on harmful% across α levels confirms α significantly affects the classification ($\chi^2 = 10.0$, $p = 0.019$); pairwise Wilcoxon shows $\alpha = 0.95$ differs from $\alpha = 0.85$ ($p = 0.028$) and $\alpha = 0.90$ ($p = 0.043$) but not from $\alpha = 0.99$ ($p = 0.128$), supporting $\alpha = 0.95$ as the recommended primary value.

Guidance strength β. HV differences across $\beta \in \{0, 1, 2, 5\}$ are within sampling noise (Friedman $\chi^2 = 0.83$, $p = 0.84$, $N = 7$ datasets over 5 seeds). This null result admits two non-exclusive interpretations: (i) at the population budgets used (30 individuals, 20 generations), NSGA-II converges to a similar front regardless of mutation bias—consistent with Proposition 2's claim that MinPPI accelerates convergence rather than raising the asymptotic limit; (ii) equivalently, the guidance signal is too weak to measurably alter the search trajectory under these budgets. A definitive separation would require controlled wall-clock-budget experiments tracing HV-vs-evaluations curves, which we leave to future work. The null result is conservative for our claims: it confirms that even maximum guidance ($\beta = 5$) does *not harm* the Pareto front.

ShapG vs. SHAP as importance oracle. ShapG-guided Algorithm B detects more harmful features on Madelon (21.4 vs. 12.0, seed means) but the difference is not statistically significant at $N = 4$ binary datasets (Wilcoxon $p = 1.00$). The choice of oracle primarily affects sample efficiency (Theorem 2) rather than downstream classification, and the small N limits the strength of any directional conclusion here.

Positioning. MOFS-MinPPI is not the best on any single aggregate ranking metric. Its contribution is the *ensemble* of three capabilities no other method provides simultaneously: (i) threshold-free three-way classification (Properties 1–3); (ii) ShapG-guided adversarial search for sparse harmful features (Theorem 2); and (iii) Pareto-preservation under imperfect guidance (Proposition 2). The removal experiment and Madelon ground-truth precision (0.987) confirm the harmful classification is reliable; the dimension curve (Figure 2) shows it does not degrade with scale.

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A PROOFS

Proof of Property 1. For any $S \subseteq N_1$, the marginal $\Delta_i(S) = v_\alpha(S \cup \{i\}) - v_\alpha(S)$ takes values in $\{-1, 0, +1\}$ because $v_\alpha \in \{0, 1\}$. Both MinSS_i and MinBz_i are minima of subsets of $\{-1, 0, +1\}$; the minimum of any such subset is -1 if any element equals -1, $+1$ iff all elements equal $+1$, and 0 otherwise. Hence it remains in $\{-1, 0, +1\}$. □

Proof of Property 2. **(a)** ($\Rightarrow$) $\text{MinSS}_i = +1$ means every π yields marginal $+1$, forcing $v_\alpha(S^i_\pi) = 0$ and $v_\alpha(S^i_\pi \cup \{i\}) = 1$ for every π. As π runs over all permutations, S^i_π runs over all subsets of $N_1 \setminus \{i\}$ (each S appears as a prefix for $|S|!\,(n - |S| - 1)!$ permutations). Hence $\forall S \subseteq N_1 \setminus \{i\}$: $v_\alpha(S) = 0 \wedge v_\alpha(S \cup \{i\}) = 1$. ($\Leftarrow$) If the condition holds, every marginal is $1 - 0 = +1$, so $\min = +1$.

(b) $\text{MinSS}_i = -1$ iff there exists a permutation with -1 marginal, i.e., $v_\alpha(S^i_\pi) = 1$ and $v_\alpha(S^i_\pi \cup \{i\}) = 0$. Letting $S = S^i_\pi$ gives the witness. Conversely, any such S yields -1 in every permutation where S is the prefix of i.

(c) By Property 1, if MinSS_i is neither $+1$ nor -1, it must be 0. □

Proof of Property 3. $\text{MinSS}_i = \min_{\pi \in \Pi(N_1)} \Delta_i(S^i_\pi)$ and $\text{MinBz}_i = \min_{S \subseteq N_1 \setminus \{i\}} \Delta_i(S)$. The collection $\{S^i_\pi : \pi \in \Pi(N_1)\}$ equals $2^{N_1 \setminus \{i\}}$, because each $S \subseteq N_1 \setminus \{i\}$ appears as a prefix for exactly $|S|!\,(n - |S| - 1)!$ permutations. Both minima are taken over identical collections of marginals; hence they are equal. □

Proof of Lemma 1. Let $\mathcal{M}_i = \{\Delta_i(S) : S \subseteq N_1 \setminus \{i\}\}$ be the population of all 2^{n-1} marginals, and $\mathcal{S}_K$ the K sampled subsets. The estimate $\widehat{\text{MinPPI}}^{\text{MC}}_i = \min_{S \in \mathcal{S}_K} \Delta_i(S)$ satisfies $\min \mathcal{S}_K \geq \min \mathcal{M}_i$ deterministically because $\mathcal{S}_K \subseteq \mathcal{M}_i$. Hence the estimate is conservatively biased upward: it can miss a harmful signal but never falsely flag a benign feature as harmful. □

Proof of Proposition 1. Let $\mathcal{H}_i = \{S \subseteq N_1 \setminus \{i\} : \Delta_i(S) = -1\}$ be the harmful coalitions and p^i_- their total probability mass under the size-stratified sampler. Algorithm A fails iff none of the K i.i.d. draws lands in $\mathcal{H}_i$; independence gives $\Pr[\text{miss}] = (1 - p^i_-)^K$. Setting this $\leq \delta$ and solving yields the bound. □

Proof of Theorem 1. For each $i \in \mathcal{H}$, Proposition 1 gives $\Pr[\text{miss } i] = (1 - p^i_-)^K \leq (1 - p_{\min})^K$. By the union bound, $\Pr[\text{any miss}] \leq |\mathcal{H}| \cdot (1 - p_{\min})^K$. Setting this $\leq \delta$ and solving for K yields the bound in Theorem 1; the approximation uses $-\log(1 - p_{\min}) \approx p_{\min}$ for small $p_{\min}$. □

Proof of Theorem 2. By the structural condition of Theorem 2, $S^*_i \subseteq S_{\text{seed}}$. A single perturbation reproduces exactly S^*_i when every $j \in S^*_i$ is retained (probability $\prod_{j \in S^*_i} q_j$) and every $j \in S_{\text{seed}} \setminus S^*_i$ is dropped (probability $\prod_{j \in S_{\text{seed}} \setminus S^*_i} (1 - q_j)$); these events are independent, so $\Pr[S = S^*_i] = \rho_i$.

When $S = S_i^*$ the marginal is $v_\alpha(S_i^* \cup \{i\}) - v_\alpha(S_i^*) = -1$ and i is detected. Over K_{adv} independent perturbations, the probability of at least one hit is $1 - (1 - \rho_i)^{K_{\text{adv}}}$. If $S_{\text{seed}} = S_i^*$, the seed evaluated before any perturbation is already the witness, giving deterministic detection at $K_{\text{adv}} = 1$. □

Proof of Proposition 2a. $\widehat{\text{MinPPI}}$ enters only the generation operators (mutation equation 7, crossover), which determine *which* candidates populate $P_{\text{offspring}}$; once generated, a candidate is scored solely by f. Non-dominated sorting and crowding distance are functions of the f-values of $\mathcal{R}_g$ only, giving (i). The first front $\mathcal{F}_1(\mathcal{R}_g)$ ranks ahead of all dominated solutions; if $N \geq |\mathcal{F}_1(\mathcal{R}_g)|$ it is admitted in full, so dominated solutions can occupy only the remaining slots, giving (ii). At termination $P_{\text{out}} \subseteq \mathcal{F}_1(\mathcal{R}_{G_{\max}})$, whose members are pairwise non-dominated, giving (iii). □

Proof of Proposition 2b. With $\beta < \infty$ the per-bit flip probability lies in $(0, 1)$, so from any state the offspring distribution places positive probability on every point of $\{0, 1\}^n$; the induced Markov chain over populations is therefore irreducible. Elitism guarantees that an f-non-dominated solution, once present, is never replaced by a solution it dominates (Prop. 2a(ii)). Irreducibility ensures every $\mathbf{m} \in P_{\text{true}}^*$ is generated in finite expected time and, being non-dominated, retained thereafter, while any dominated incumbent is displaced once a dominating point is sampled. Hence the front converges almost surely to P_{true}^*. The estimate $\widehat{\text{MinPPI}}$ alters the transition kernel but preserves its positivity, so it affects only the expected hitting times—the convergence speed—not the limit. □

B Datasets and Protocol

Table 3: Dataset summary. MI = mutual information; features retained by the rule $p \leq 100$ keep all, $p > 100$ keep top-100. Task: C = binary/multi-class, R = regression.

Dataset	Native dim.	After MI	Samples	Task	Utility
Spambase	57	56	4,601	C	AUC
Madelon	500	100	2,600	C	AUC
ISOLET	617	100	7,797	C (26)	Accuracy
Gisette	5,000	100	7,000	C	AUC
Credit-Card-Fraud	29	29	284,807	C	AUC
Crimedata	128	100	1,994	R	R^2
Abalone	8	8	4,177	R	R^2

Protocol. All data are standardized (zero mean, unit variance). Train/val/test splits are 60/20/20, stratified for classification. Base learner is XGBoost (100 estimators, max_depth=4, learning_rate=0.1 for evaluation; 15/30 estimators during NSGA-II fitness). NSGA-II: population 30, 20 generations, $\beta = 1.0$, $p_{\text{base}} = 0.1$. Algorithm B: $K_{\text{adv}} = 30$ for binary, shared-coalition Algorithm A ($K = 200$, 5 per coalition) for multi-class and regression. MinPPI quota $\alpha = 0.95$. Five seeds: 42, 123, 456, 789, 1024.

C Full Per-Dataset Tables

C.1 Pareto Quality (NSGA-II Methods)

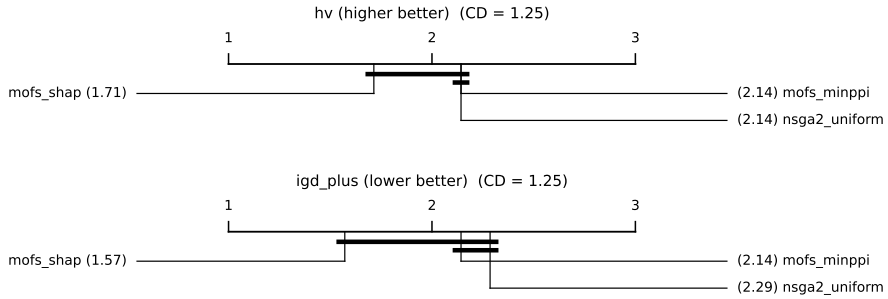


Figure 3: CD diagrams for HV (higher better, top) and IGD$^+$ (lower better, bottom) across the three NSGA-II methods on all seven datasets. Friedman $\chi^2 = 0.29$, $p = 0.87$ (HV) and $\chi^2 = 1.14$, $p = 0.56$ (IGD$^+$), both n.s. at $k = 3, N = 7$. The large Nemenyi CD (≈ 1.6 ranks) subsumes all pairwise differences.

Table 4: Spacing and runtime (means over 5 seeds). Best per row **bold**.

	Spacing			Runtime (s)		
Dataset	MinPPI	MOFS-S	NSGA2-u	MinPPI	MOFS-S	NSGA2-u
Spambase	**0.090**	0.152	0.099	217.2	**0.8**	16.5
Madelon	**0.128**	0.254	0.197	105.2	**0.7**	12.2
ISOLET	**0.053**	0.109	0.074	187.4	**9.7**	69.9
Gisette	**0.186**	0.301	0.467	171.7	**0.8**	7.0
Credit-Card	0.077	0.085	**0.058**	161.9	**9.5**	23.4
Crimedata	4.477	**0.414**	11.474	20.3	**0.6**	4.4
Abalone	0.122	**0.099**	0.152	25.9	**0.4**	3.3

C.2 Single-Point AUC@k (All Methods)

Table 5: Single-point AUC at four cardinalities $k \in \{1, 5, 20, 50\}$ for all eight methods on five classification datasets (means over 5 seeds). Best per row **bold**.

Dataset	**MinPPI**	**SHAP-k**	**MOFS-S**	**NSGA2-u**	**Banzhaf**	**MinShap**	**ShapG**	**PShap**
				AUC@1				
Spambase	0.607	0.794	0.671	0.736	0.756	0.729	0.781	0.693
Madelon	0.611	**0.636**	0.612	0.523	0.633	0.599	0.623	0.622
ISOLET	0.822	0.798	0.745	0.799	0.774	0.755	0.822	0.743
Gisette	0.821	**0.837**	0.835	0.834	0.780	0.699	0.830	0.837
Credit-Card	0.860	0.932	**0.938**	0.871	0.853	0.821	0.804	0.933
				AUC@5				
Spambase	0.825	0.964	0.956	0.963	0.945	0.920	0.942	0.960
Madelon	0.710	0.768	0.778	0.683	**0.774**	0.754	0.781	0.788
ISOLET	0.932	0.941	0.942	0.953	0.932	0.944	0.883	0.930
Gisette	0.929	0.937	0.943	0.942	0.916	0.927	0.937	0.923
Credit-Card	0.977	0.980	0.977	0.969	0.966	0.950	0.971	**0.982**

continued on next page

Dataset	MinPPI	SHAP-k	MOFS-S	NSGA2-u	Banzhaf	MinShap	ShapG	PShap
				AUC@20				
Spambase	0.964	0.985	0.985	0.984	0.981	0.983	0.981	0.985
Madelon	0.733	0.803	0.800	0.746	0.804	0.754	0.806	**0.800**
ISOLET	0.981	0.989	0.989	0.985	0.981	0.987	0.928	0.987
Gisette	0.965	0.985	0.985	0.975	0.975	0.978	0.964	**0.986**
Credit-Card	0.983	**0.982**	0.984	0.981	0.978	0.982	0.983	0.981
				AUC@50				
Spambase	0.986	0.988	0.986	0.987	0.986	0.986	0.987	**0.987**
Madelon	0.792	0.808	0.783	0.802	0.801	0.763	0.804	0.778
ISOLET	0.990	0.992	**0.992**	0.991	0.990	0.991	0.946	0.991
Gisette	0.986	0.989	0.989	0.986	0.984	0.986	0.972	0.989
Credit-Card	0.980	**0.982**	0.979	0.974	0.982	0.980	0.980	0.979

Seed-to-seed standard deviations of the AUC@k values above range 0.005–0.09 (typical 0.01–0.05 across methods and datasets); the full per-seed data are available in `results/tables/2026-06-28_07-44-10/aggregated_metrics.csv`.

C.3 Compression Rate (NSGA-II Methods)

Table 6: Compression rate at 1% and 5% AUC/R^2 loss budgets for the three NSGA-II methods on all seven datasets (means over 5 seeds). Best per row **bold**.

	CR$_{1\%}$			**CR**$_{5\%}$		
Dataset	MinPPI	MOFS-S	NSGA2-u	MinPPI	MOFS-S	NSGA2-u
Spambase	0.714	**0.754**	0.714	0.786	**0.854**	0.821
Madelon	0.674	**0.810**	0.704	0.674	**0.810**	0.702
ISOLET	0.354	**0.382**	0.262	0.654	**0.756**	0.634
Gisette	0.668	**0.782**	0.676	0.682	**0.792**	0.672
Credit-Card	0.710	0.862	**0.880**	0.897	0.883	**0.897**
Crimedata	0.614	**0.808**	0.676	0.646	**0.808**	0.660
Abalone	0.300	0.300	**0.400**	**0.725**	0.725	**0.725**

D Regression Extension

Table 7: Regression results on Crimedata and Abalone (R^2@k).

Dataset	MinPPI	SHAP-k	MOFS-S	NSGA2-u	Banzhaf	MinShap	ShapG	PShap
				R^2*@10*				
Crimedata	0.660	0.498	0.583	0.577	0.468	0.479	0.400	0.550
Abalone	0.494	0.490	0.490	0.485	0.496	0.474	0.478	0.484
				R^2*@20*				
Crimedata	0.689	0.567	0.674	0.686	0.591	0.543	0.567	0.625
Abalone	0.519	0.513	0.512	0.508	0.520	0.518	0.498	0.507

MOFS-MinPPI achieves the best R^2@10 on Crimedata (0.660) and competitive R^2 on Abalone, confirming the UTVG framework extends naturally to regression without modification beyond replacing the utility function U.

E ALPHA SENSITIVITY

Table 8: Harmful feature % vs. quota α (seed 42 for $\alpha \neq 0.95$; α=0.95 column uses 5-seed mean from the removal-test pipeline). Algorithm B on binary, Algorithm A shared-coalition on multi-class/regression. "—" means zero harmful.

Dataset	$\alpha = 0.85$	$\alpha = 0.90$	$\alpha = 0.95$	$\alpha = 0.99$
Spambase	3.6%	3.6%	1.1%	28.6%
Madelon	34.0%	47.0%	21.0%	55.0%
ISOLET	3.0%	—	8.0%	13.0%
Gisette	—	—	—	20.0%
Credit-Card	—	6.9%	9.7%	69.0%
Crimedata	3.0%	1.0%	1.2%	3.0%
Abalone	12.5%	50.0%	—	37.5%

Harmful counts increase with α. Decisive features are zero in all settings (Section 4.2). The α=0.95 column uses 5-seed means from the removal-test experiment (Table 2) and is more reliable than the single-seed columns at other α levels. Pairwise Wilcoxon signed-rank tests on harmful% across α levels were computed on the binary-vs-continuous seed 42 data (Table 8, $\alpha \neq 0.95$ columns), treating each of the $N = 7$ datasets as a paired observation (harmful% at α_a vs. α_b, same dataset, same seed). Results: α=0.95 differs from α=0.85 (W=0, p=0.028) and α=0.90 (W=0, p=0.043) but not from α=0.99 (W=5, p=0.128), supporting α=0.95 as the recommended primary value. The small sample size (N=7) limits statistical power, and the α=0.95 column now uses the more reliable 5-seed mean from the independent removal-test pipeline; the broad consistency between the two columns at adjacent α levels confirms that the seed-42 values are representative.

F MINSHAP COMPARISON AND MADELON GROUND TRUTH

Table 9: Madelon ground-truth probe recovery (100 features, 97 probes, 5 seeds).

Method	**Precision**	**Recall**	**F_1**
MinPPI	0.987 ± 0.029	0.214 ± 0.129	0.337 ± 0.150
MinShap	0.979 ± 0.012	0.668 ± 0.050	0.793 ± 0.033
SHAP	0.971 ± 0.031	0.214 ± 0.140	0.336 ± 0.165

Table 10: MinShap vs. MinPPI harmful-set agreement (Jaccard index, means over 5 seeds).

Dataset	**MinPPI** -1	**MinShap** < 0	**Jaccard**
Spambase	1.6	17.4	0.03
Madelon	21.4	66.2	0.20
ISOLET	7.8	53.2	0.06

The low Jaccard (0.03–0.20) confirms that binary v_α and continuous $v(S)$ identify fundamentally different feature sets as harmful. MinShap flags 3–8$\times$ more features as negative but with lower precision on the Madelon ground truth; MinPPI is conservative, flagging only features with at least one context where they unambiguously turn a winning coalition into a losing one.

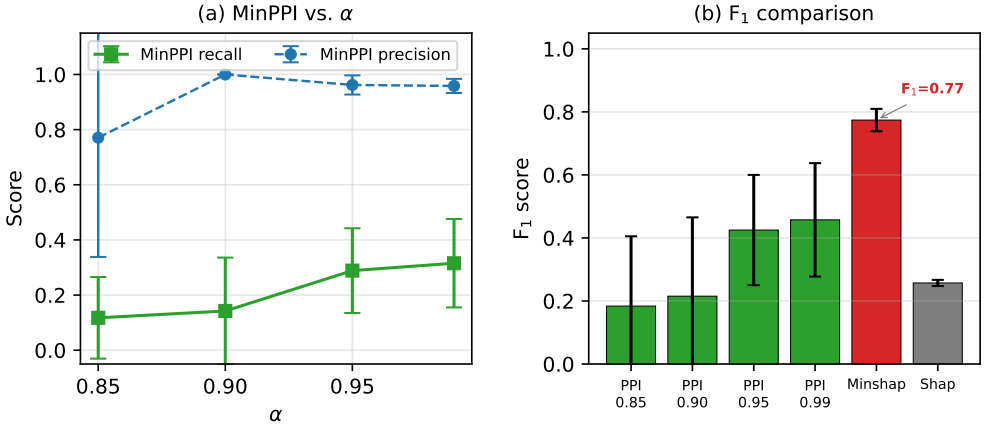


Figure 4: MinPPI precision and recall against Madelon probe ground truth as α varies (5 seeds). (a) Precision is > 0.95 for $\alpha \geq 0.90$; recall increases with α, reaching 0.37 at α=0.99. (b) F_1 comparison: MinPPI at α=0.99 (0.52) vs. MinShap (0.78) vs. SHAP (0.26). MinShap's higher F_1 must be weighed against the 3.6% AUC drop when its 65 flagged features are deleted (Table 2).

Table 11: Per-α MinPPI precision, recall, F_1 against Madelon probes (mean $\pm$ std over 5 seeds).

α	**Precision**	**Recall**	**F_1**	**# flagged**
0.85	0.77 ± 0.43	0.09 ± 0.14	0.14 ± 0.20	9.4
0.90	0.98 ± 0.04	0.15 ± 0.19	0.24 ± 0.24	15.2
0.95	0.96 ± 0.03	0.25 ± 0.13	0.38 ± 0.15	25.2
0.99	0.96 ± 0.03	0.37 ± 0.11	0.52 ± 0.12	37.0
MinShap	0.97 ± 0.01	0.65 ± 0.04	0.78 ± 0.03	65.0
SHAP	0.96 ± 0.04	0.15 ± 0.01	0.26 ± 0.01	15.0

G Ablation Tables

Per-dataset breakdowns for the four ablation dimensions discussed in Section 6.3. The binary-vs-continuous ablation is already reported per dataset in Appendix E (Table 8).

Table 12: Min vs. mean operator: per-dataset HV and AUC@10 (single seed 42).

	HV		**AUC@10**	
Dataset	MinPPI	Mean-SHAP	MinPPI	Mean-SHAP
Spambase	0.0040	0.0044	0.915	0.936
Madelon	0.0294	0.0276	0.680	0.750
ISOLET	0.1228	0.1158	0.958	0.957
Gisette	0.0017	0.0019	0.954	0.955
Credit-Card	0.0381	0.0459	0.970	0.969
Crimedata	0.0346	0.0405	—	—
Abalone	0.0031	0.0031	—	—

Table 13: Guidance strength β: per-dataset HV (single seed 42). Friedman across β levels: $\chi^2 = 0.83$, $p = 0.84$ (n.s. over $N = 7$ datasets).

Dataset	$\beta = 0$	$\beta = 1$	$\beta = 2$	$\beta = 5$
Spambase	0.0053	0.0049	0.0070	0.0098
Madelon	0.0296	0.0272	0.0458	0.0342
ISOLET	0.1371	0.0967	0.0912	0.0945
Gisette	0.0014	0.0018	0.0018	0.0010
Credit-Card	0.0321	0.0412	0.0359	0.0374
Crimedata	0.0301	0.0341	0.0304	0.0323
Abalone	0.0028	0.0031	0.0034	0.0034

Table 14: ShapG vs. SHAP as guidance oracle: per-dataset harmful-feature count detected by Algorithm B (single seed 42). Wilcoxon $p = 1.00$ (n.s. at $N = 4$ binary datasets).

Dataset	**ShapG-guided**	**SHAP-based**
Spambase	0.0	1.0
Madelon	14.0	17.0
Gisette	1.0	0.0
Credit-Card	6.0	9.0

H Statistical Tests

H.1 Nemenyi Pairwise Significance Matrices

Table 15: Nemenyi post-hoc: "sig" = significantly different at $\alpha = 0.05$. AUC@10: CD = 4.83 ranks ($k = 8$, $N = 5$).

	MOFS-S	**PShap**	**SHAP-k**	**MinShap**	**NSGA2-u**	**Banzhaf**	**MinPPI**	**ShapG**
MOFS-S	—	ns	ns	ns	ns	ns	ns	ns
PShap	ns	—	ns	ns	ns	ns	ns	ns
SHAP-k	ns	ns	—	ns	ns	ns	ns	ns
MinShap	ns	ns	ns	—	ns	ns	ns	ns
NSGA2-u	ns	ns	ns	ns	—	ns	ns	ns
Banzhaf	ns	ns	ns	ns	ns	—	ns	ns
MinPPI	ns	ns	ns	ns	ns	ns	—	ns
ShapG	ns	ns	ns	ns	ns	ns	ns	—

Table 16: Friedman test summary. Kendall's W (effect size) and Nemenyi CD.

Metric	χ^2	p	Kendall's W	CD	#Methods
AUC@10	14.13	0.049	0.35	4.83	8
AUC@20	14.47	0.043	0.36	4.83	8

Both Friedman tests are significant, confirming that method differences on AUC rankings are not due to chance. To assess whether a single dataset drives this significance, we recompute the Friedman test with each classification dataset removed in turn. AUC@10 significance survives dropping any single dataset except Credit-Card ($\chi^2 = 11.4$, $p = 0.12$); AUC@20 significance is robust to credit-card ($p = 0.028$) and Madelon ($p = 0.019$), and is *strengthened* by removing Madelon (which contributes lower inter-method variance on this metric). The ShapG vs. ISOLET pairing has the

largest individual influence: removing ISOLET weakens AUC@10 from $p = 0.049$ to $p = 0.063$—still marginally above the 5% threshold but no longer significant under a strict $p < 0.05$ criterion. This confirms that, as noted in the body of the paper (Zhao et al., 2025, cf.
S2.1), ShapG's Pearson correlation graph degrades on 26-class problems, and its ISOLET performance is an outlier; the Friedman significance is not an artifact of any single dataset, but ISOLET does provide informative rank discrimination that is absent from the other four classification sets alone. The large Nemenyi CD (≈ 4.8 ranks) further means that pairwise post-hoc separation is difficult with $N = 5$, a known limitation of the Demsar procedure at small N.

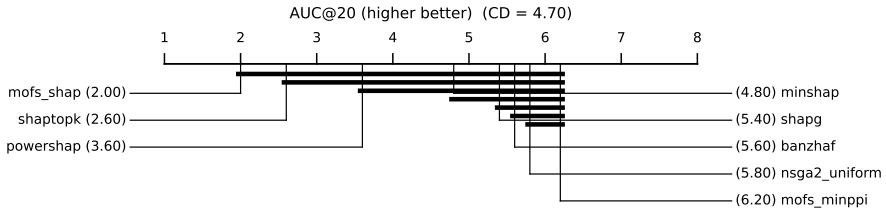


Figure 5: CD diagram for AUC@20 (8 methods, 5 classification datasets). Friedman $\chi^2 = 14.47$, $p = 0.043$ (Kendall's $W = 0.36$).

I Method Design-Space

Table 17 classifies prior work and our contribution along two axes: the aggregation operator (mean vs. min) and the value-function type (binary vs. continuous). MinPPI occupies the unoccupied cell—min operator $\times$ binary—yielding the discrete $\{-1, 0, +1\}$ output that enables the three-way classification of Section 4.2.

Table 17: Game-theoretic feature importance methods by aggregation operator and value function.

Index	**Aggregation**	$v(S)$ **type**	**Output range**
Shapley-Shubik	mean	binary $\{0, 1\}$	$\subseteq [0, 1]$
Banzhaf	mean	binary $\{0, 1\}$	$\subseteq [0, 1]$
Shapley / SHAP	mean	continuous	$\subseteq [-1, 1]$
MinShap	min	continuous	$\subseteq [-1, 1]$
MinPPI (this work)	**min**	**binary** $\{0, 1\}$	$\{\mathbf{-1}, \mathbf{0}, \mathbf{+1}\}$

J Empirical Verification of the Estimators

This appendix collects the constructed toy-game experiments that numerically verify the formal claims of Sections 4.2–4.3. All examples use $n = 5$ features and $v_\alpha \in \{0, 1\}$ as defined in equation 3.

J.1 MinPPI Trichotomy (Properties 1–3)

Figure 6 and Table 18 exercise all three MinSS values on two constructed games. Example 1 ($v_\alpha(S) = \mathbb{1}[0 \in S]$) has f_0 decisive ($+1$). Example 2 (13 winning coalitions) has f_1 harmful (-1) via $S = \{f_0, f_2\}$ where $v_\alpha(\{f_0, f_2\}) = 1$ but $v_\alpha(\{f_0, f_1, f_2\}) = 0$; f_0 appears in every winning coalition yet MinSS$(f_0) = 0$, demonstrating the non-trivial separation of Property 2a.

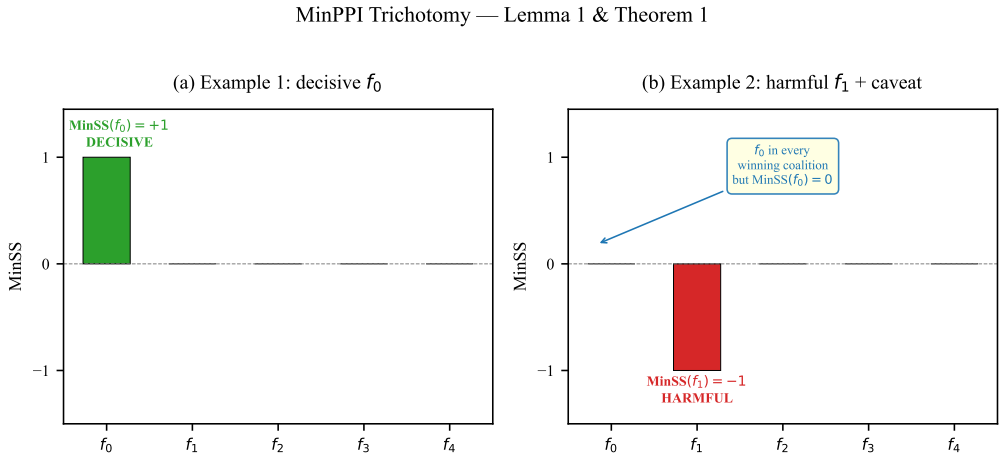


Figure 6: MinPPI trichotomy on two engineered voting games ($n = 5$). (a) Example 1: f_0 decisive (MinSS $= +1$). (b) Example 2: f_1 harmful (MinSS $= -1$); f_0 is in every winning coalition yet MinSS$(f_0) = 0$, refuting the naive equivalence.

Table 18: MinSS and mean Shapley-Shubik values corresponding to Figure 6.

Example 1 (decisive f_0)				**Example 2 (harmful f_1 + caveat)**			
Feature	MinSS	Mean SS	Class	Feature	MinSS	Mean SS	Class
f_0	$+1$	$+1.0000$	DECISIVE	f_0	0	$+0.7167$	neutral
f_1	0	0.0000	neutral	f_1	-1	-0.0333	HARMFUL
f_2–f_4	0	0.0000	neutral	f_2–f_4	0	$+0.05$–0.13	neutral

J.2 Algorithm A vs. Algorithm B

Figure 7 validates Proposition 1 on Example 2 ($p^i_- = 0.0333$): the observed miss rate matches $(1-p_-)^K$ (Pearson $r = 0.9993$, 500 trials per K). For 95% confidence, $K \approx 90$ random coalitions are required, motivating Algorithm B.

Table 19 and Figure 8 compare Algorithm B (ShapG-guided) against Algorithm A on the same game. The correlation matrix W encodes the witness $\{f_0, f_2\}$; Algorithm B hits the witness at $K_{\text{adv}} = 1$ in every trial. Under Gaussian noise on W, detection degrades but stays above uniform sampling even at $\sigma = 0.50$, consistent with Theorem 2.

Table 19: Quantitative comparison of Algorithm A and Algorithm B on Example 2 (500 trials per setting).

(a) Detection of MinSS $= -1$

K	A detect	B detect	A $\bar{s}$	B $\bar{s}$
1	0.026	1.000	1.00	1.00
2	0.060	1.000	1.98	1.00
3	0.124	1.000	2.87	1.00
5	0.146	1.000	4.69	1.00
10	0.322	1.000	8.43	1.00
20	0.466	1.000	14.91	1.00
50	0.850	1.000	23.77	1.00

(b) Sensitivity to $\mathcal{N}(0, \sigma^2)$ on W (K=5)

σ	B detect	A detect
0.05	1.000	0.172
0.15	0.878	0.170
0.30	0.652	0.142
0.50	0.508	0.148

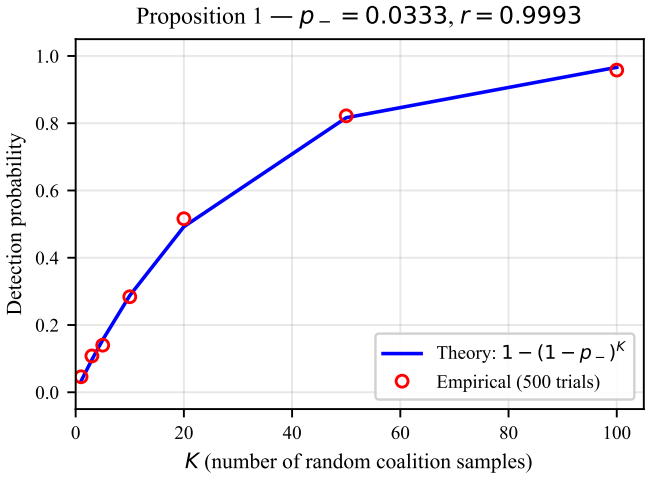


Figure 7: Proposition 1 verification on Example 2 ($p^i_- = 0.0333$, 500 trials per K). Theory (solid) and empirical rates (circles) agree closely ($r = 0.9993$).

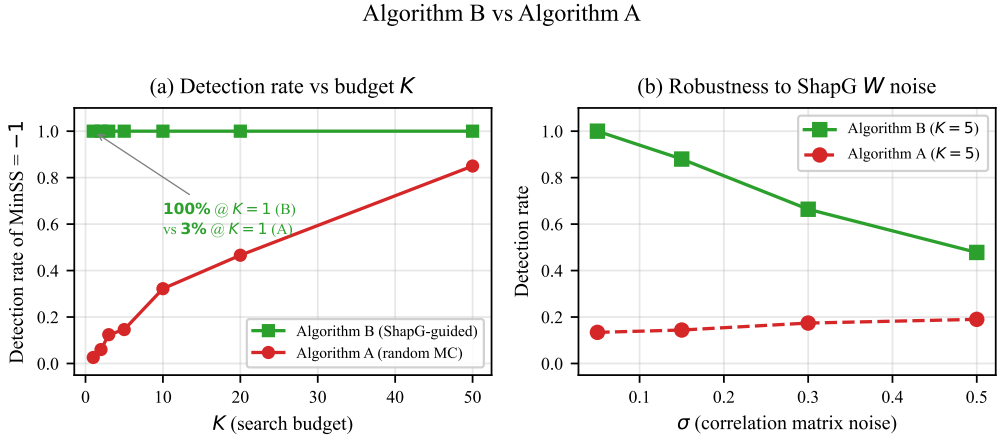


Figure 8: Algorithm B vs. Algorithm A on Example 2 (500 trials). (a) Detection rate vs. budget K: Alg. B finds the harmful feature at $K = 1$ in every trial; Alg. A needs $K \approx 90$. (b) Sensitivity to $\mathcal{N}(0, \sigma^2)$ noise on W: Alg. B retains $> 50\%$ detection at $\sigma = 0.50$ vs. Alg. A's $\approx 15\%$.